

MUHAMMAD ZIA UL HASSAN

Mobile no: +92-317-4694078 Email: mzia9612@gmail.com

<https://www.linkedin.com/in/m-zia-ul-hassan-8076a7206/>

Educational Background

University of Engineering and Technology (UET), Lahore, Pakistan

B.Sc. Mechatronics and Control Engineering

Oct 2020 – Aug 2024

Major Courses: Robotics, Embedded Systems, Machine Learning, Control Systems, Computer Programming, *Industrial Automation, Signals and Systems and Discrete Control Systems*

Academic Experience

HOME AUTOMATION

Jan– May 2021

- **Electronics:** Designed an automated door lock, fan controls, and lighting systems using IoT technology, enhancing home safety, comfort, and energy efficiency.
- Sensing using micro-controller **TIVA**, respective sensors, and developed its algorithm in **C++** language.

PNEUMATIC PICK AND DROP

Nov 2022 – Jan 2023

- **Automation:** Developed a pneumatic pick-and-drop system using relay logic. This system efficiently automates the process of picking up and placing objects, enhancing productivity and precision in industrial applications.

HAND SIGNS DETECTOR

Apr – May 2022

- **Programming:** Created a hand signs detector in Python using Tkinter, OpenCV, and MediaPipe. This system accurately identifies and interprets various hand gestures, enabling intuitive human-computer interaction for accessibility and control applications.

AIR FLOW BALL BALANCING

Dec 2021 – Jan 2022

- **Control:** Designed an air flow ball balancing system using advanced control systems. This setup maintains the ball's position in mid-air by dynamically adjusting airflow, demonstrating principles of feedback control and stability.

PNEUMATIC PICK AND DROP USING PLC

Nov 2022 – Jan 2023

- **Automation:** Developed a pneumatic pick-and-drop system using PLC technology. This automated solution enhances precision and efficiency in handling tasks, making it ideal for industrial applications.

MULTIFEATURED ROBOT: MEET AND GREET II (FYP)

Aug 2023 – May 2024

- **Machine Learning, Robotics and Automation:** My Final Year Project (FYP) focused on designing a Sociable Robot with advanced capabilities including Human-Robot Interaction (HRI), Voice Activation, Emotional Recognition, and comprehension of American Sign Language (ASL). Leveraging 3D CAD modeling, the project aimed to create an empathetic and inclusive robot capable of engaging in meaningful social interactions with humans, fostering companionship and accessibility through innovative technology.

Technical Skills:

- Programming (C, C++, Python, PLC),
- Embedded Systems
- Robotics
- Modeling and Simulation,

- Machine Learning
- Industrial Automation

Software Skills:

| Matlab | ANSYS | Simulink | SolidWorks | EasyEda | Microsoft Visual Studio,
 | Microsoft Office | Open PLC Editor | AutoCAD | Micro Vision Keil | Energia IDE
 | Proteus | TIA Portal | Canva

Professional Experience

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- **Automated Bottle Filling System (Next Cola Pvt. Ltd.):** **Jun-Aug 2022**
 Worked on an automated bottle filling system at **Next Cola Pvt. Ltd.**, using **Siemens TIA Portal** for PLC programming. Integrated sensors and actuators, implemented control logic, and supported HMI setup and troubleshooting to enhance system reliability.
 - **Overseas:** **Jun – Aug 2023**
 - ***Trainee Sales and Application Engineer:*** As a Trainee Sales and Application Engineer, I gained invaluable experience at an overseas enterprise, developing technical sales skills and understanding customer needs. I collaborated with diverse teams to support product applications, enhancing my problem-solving and communication abilities in an international business environment.
 - **Diesel Generator ATS Panel Simulation (PLC-Controlled):** **Feb-May 2024**
 Designed and simulated a PLC-based **Automatic Transfer Switch (ATS)** system using **Siemens TIA Portal** to automate load transfer between utility and diesel generator. Developed ladder logic with interlocks, timers, and fault handling for safe switching and auto-retransfer on mains recovery.
 - **Automation Intern – Basfa Textile:** **Jun-Aug 2024**
 Completed an automation internship at **Basfa Textile**, working with **Mitsubishi FX Series** and **Siemens S7-1200 PLCs**. Gained hands-on experience in ladder logic development, sensor/actuator integration, HMI support, I/O testing, and basic panel wiring using **GX Developer** and **TIA Portal**.
 - **Punjab Group of Colleges (City Campus, Lahore)** **Sep 2024 – Now**
 - ***Head of Robotics and AI:*** Directed robotics and automation programs, designing hands-on curricula and optimizing lab operations to elevate student skills in control systems and practical automation. Spearheaded cross-departmental projects, fostering a collaborative environment that encouraged innovative problem-solving. Enhanced student engagement and academic outcomes, equipping students with the foundational and technical skills for advanced roles in robotics and engineering.

Extracurricular and Volunteer Experience

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- BLOOD DONOR SOCIETY SILLANWALI** **Jan 2020 – Present**
- ***Volunteer:*** Volunteered as a blood donor finder at the Blood Donor Society Sillanwali, connecting donors with those in need and ensuring timely blood donations. My efforts contributed to saving lives and supporting community health initiatives.

CLICK AND HELP**Jan 2021 – Present**

- **Volunteer:** I volunteered at Click and Help, a society dedicated to assisting orphans, widows, and those in need. My work involved supporting various initiatives to improve their quality of life and provide essential resources, making a meaningful impact on the community.

ROSHAN KAL ACADEMY:**Aug – Sep 2024**

- Awarded for successful completion of 10 intensive modules, demonstrating a strong commitment to growth and continuous learning. This achievement provides a solid foundation in skills and knowledge for future pursuits.

DAST E KHAIR FOUNDATION:**Dec 2024 – April 2025**

- Actively working as Director of Finance, initiatives to support orphans, widows, and underprivileged individuals. Contributed to providing essential resources and implementing programs focused on improving their quality of life.

HOBBIES

- Reading, Writing, Poetry: Expressive and creative through literature and verse.
 - Gardening: Find solace and joy in nurturing plants.
 - Traveling, Photography: Enthusiastic explorer and storyteller.
- Critical Thinking: Analytically evaluate information for informed decision making.